MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NUMERICAL INDICATOR

TYPE 7977

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Gas filled, cold cathode

See figure 1

Mounting position: See note 8

Weight: 0.2-ounce nominal

ABSOLUTE RATINGS:

Parameter:	Ebb	Ik (individual)	Rp	TA	Barometric pressure, reduced
Unit:	Vdc	mA	Ohms	°C	mmHg
Maximum:	---	2.0	---	+55	---
Minimum:	170	---	---	-20	35
TEST CONDITIONS:	170	---	15,000 ±1%	---	---

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1						
3347	Ionization voltage (1) (with illumination)	-	Ambient illumination = 5 to 50 ftC	Ebb	---	170	Vdc
3333	Uniformity of cathode glow	7	Ambient illumination = 5 to 50 ftC	---	---	---	---
1256	Electrode current (cathode)	7	Individual cathodes	Ik	0.7	1.4	mA
	Quality conformance inspection, part 2						
1031	Variable-frequency vibration	3,9		---	---	---	---
---	Shock	9,10	No voltages applied; 250 G; duration = 1.5 ±1 ms	---	---	---	---
1002	Barometric pressure, reduced	6	eb = 170; Rp = 15,000 ohms; pressure = 35 mmHg	---	---	---	---
---	Light output	5,7	Light individual cathodes	---	0.15	---	ftC
---	Alinement, digits	2		---	---	±5	Degrees
(B) 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test (1)	1	Group C; sequential counting; 1 Hz; t = 1,000 hours	---	---	---	---
---	Life-test (1) end points:						
3333	Uniformity of cathode glow	7		---	---	---	---
1256	Electrode current (cathode)	7		Ik	0.4	1.7	mA
---	Light output	-		---	0.15	---	ftC
---	Life-test (2)	4	Group C; one numeral cathode glowing steadily; t = 1,000 hours	---	---	---	---
---	Life-test (2) end points:						
3333	Uniformity of cathode glow	7	Numeral under test	---	---	---	---
1256	Electrode current (cathode)	7	Numeral under test	Ik	0.4	1.7	mA
---	Light output	-	Numeral under test	---	0.15	---	ftC
(B) ---	Cathode	11	Glow discharge (neon)	---	---	---	---

NOTES:

1. The tube shall be energized with a duty cycle greater than 90 percent.
2. Centerline of numerals, drawn through the top of the numeral support posts, shall be within the tolerance specified herein in relation to line drawn on the base of the tube through pins 1 and 7.
3. On each plane the 10 cathodes shall be sequentially energized at the test conditions.

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NOTES: -Continued

4. Group C, life-test sampling plan, and Group C, establishment of prerelease eligibility (MIL-E-1 paragraphs) should be used with the following exceptions:

   (a) Group C life-test sampling plan.

      (1) The first sample size = 3.
      (2) The first cumulative sample size = 3.
      (3) The second sample size = 3.
      (4) The second cumulative sample size = 6.

   (b) Group C, establishment of prerelease eligibility.

      (1) A minimum of 15 tubes shall-------etc.
      (2) Replace all reference to the current 4-tube sample with current 3-tube sample. Replace all reference to the last 16 with the last 15.

   (c) One tube shall have the top numeral cathode glowing steadily. One tube shall have the bottom numeral cathode glowing steadily. One tube shall have the "0" numeral cathode glowing steadily.

5. Light output of each numeral is measured with a footcandle (ftC) meter and an eye-corrected Photronic cell, or equivalent. The entire cell area shall be illuminated and shall be coupled to the tube by a light-tight cylinder so that the face of the cell is 1-1/2 inches (38.10 mm) from the tube base.

6. The tube shall be tested by using a 60 Hz sine wave, full-wave rectifier and unfiltered, having the indicated peak value.

7. No glow shall be permitted on any tube part other than the numeral under test.

8. The tube socket shall be mounted with respect to the viewing position so that a line intersecting pins 1 and 7 is vertical with pin 7 on top. The numbers are viewed through the top of the tube.

9. No permanent shorts or open circuits will be permitted.

10. Shock test shall be made on any shock machine capable of producing a half-sine wave shock waveform of the specified duration and amplitude. No voltage will be applied during this test. Each tube shall be subjected to a total of 20 shocks, that is, 5 shocks in each of positions X, Y, Z plus, and Z minus in any sequence. This is a destructive test. Tubes subjected to the shock test shall not be delivered on the contract or order.

(8) 11. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 17, 80
DSA - ES

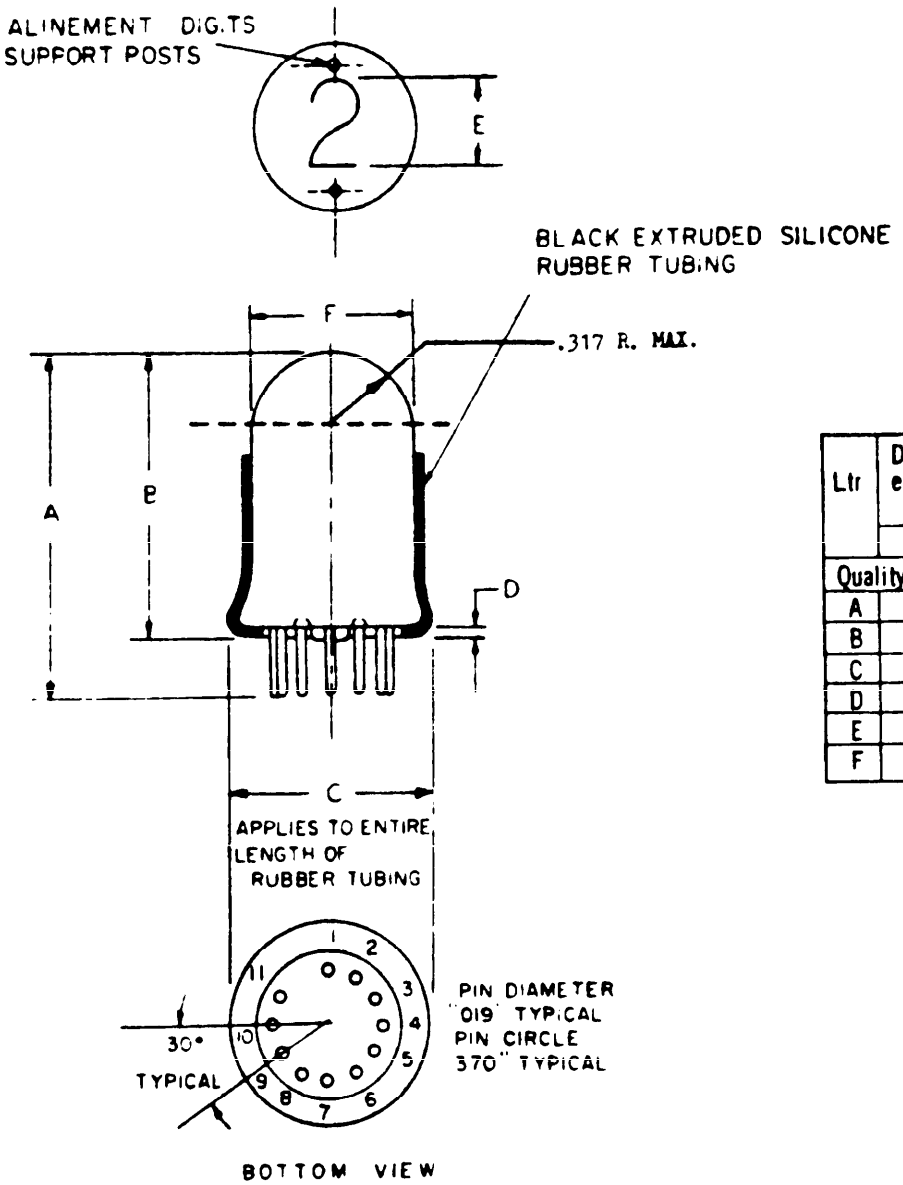
User activities:
Navy - AS, OS, MC, CG
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Agent:
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	1.000 (25.40)	1.200 (30.50)
B	.840 (21.34)	.940 (23.88)
C	---	.680 (17.27)
D	---	.064 (1.63)
E	.295 (7.49)	.315 (8.00)
F	---	.635 (16.13)

PIN CONNECTIONS	
Pin No.	Element
1	k1
2	k2
3	k3
4	k4
5	k5
6	k6
7	k7
8	k8
9	k9
10	k0
11	a

1234567890

NUMERAL DESIGN

FIGURE 1. Outline drawing of electron tube type 7977.

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WASHINGTON, D. C. 20360

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